PHARMATRACK: PHARMACY INVENTORY AND TRANSACTION MANAGEMENT SYSTEM

A System Proposal Presented to the Faculty of the ICT

Senior High School Unit Basic Education Department

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In Partial Fulfillment of the Academic

Requirements for the Subject

VISUAL PROGRAMMING

(VP)

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EXECUTIVE SUMMARY

PharmaTrack is a comprehensive pharmacy management solution purposely designed to automate and streamline the distribution and tracking of branded, generic, and over-the-counter medicines. Recognizing that many small to medium-sized pharmacies continue to rely on manual record-keeping, this system directly addresses recurring operational problems such as inaccurate inventory counts, inadequate expiration date monitoring, and delays in generating sales reports. The primary aim of this project is to replace these inefficient manual processes with an integrated digital system that ensures both operational accuracy and pharmaceutical safety.

During the System Analysis and Design phase, the developers identified major bottlenecks in existing inventory and transaction handling workflows. To address these issues, the PharmaTrack: Pharmacy Inventory and Sales Management System was proposed and developed. The system incorporates automated sales computation, real-time stock level monitoring, expiration date alerts, and a secure data storage framework supported by a MySQL relational database.

Through Visual Programming, the development team leveraged Java Object-Oriented Programming (OOP) to construct an intuitive user interface that reduces manual input and minimizes human error. After undergoing rigorous Black Box Testing procedures, the system was confirmed to satisfy all defined functional requirements and meet user expectations. A structured maintenance plan was also established to ensure the system's long-term reliability, performance, and continued relevance to pharmacy operations.

ACKNOWLEDGMENT

The developers of PharmaTrack sincerely extend their deepest gratitude to the following individuals and organizations whose invaluable contributions made the successful completion of this project possible:

To the Almighty God, for the wisdom, strength, patience, and perseverance generously provided throughout every challenge encountered during the planning, development, and testing of this system.

To the project's dedicated Adviser, for the consistent guidance, technical expertise, and constructive feedback throughout the semester. The lessons on Visual Programming and Database Management Systems served as the solid foundation upon which this project was built.

To the Philippine Women's College of Davao, for providing a conducive and supportive learning environment, as well as the necessary laboratory facilities that enabled the developers to build, test, and refine the software effectively.

To our Parents and Guardians, for their unwavering moral and financial support, and for understanding the long hours and dedication required to bring this project to completion.

Finally, to our teammates and peers, for the collaborative effort, shared commitment, and team spirit that drove PharmaTrack from concept to a fully functional and reliable system.

DEDICATION

This work is wholeheartedly dedicated to our Parents and Guardians, whose unconditional love, unwavering support, and financial sacrifices have made it possible for us to pursue our education and bring this project to fruition.

To our ICT Teachers and Mentors, for continuously inspiring us to think critically, code creatively, and strive for excellence in the field of information and communications technology.

To the Future ICT Students of Philippine Women's College of Davao, may this system serve as a meaningful reference and source of inspiration as you embark on your own journey in software development.

And above all, to the Almighty God, the ultimate source of our strength, intellect, and perseverance, to whom we offer all the glory of this accomplishment.

TABLE OF CONTENTS

EXECUTIVE SUMMARY	i
ACKNOWLEDGMENT	ii
DEDICATION	iii
TABLE OF CONTENTS	iv
LIST OF TABLES	vi
LIST OF FIGURES	vii
PART I SYSTEM ANALYSIS REPORT	
Introduction	1
General Objective	2
Specific Objectives	2
Data Flow Diagrams	3
Activity Diagrams of the System	3
Class Diagram of the System	7
Context Level Data Flow Diagram	8
Level 0 Data Flow Diagram	9
Functional Requirements	9
Use Cases	17
Use Case Diagram	21
Gantt Chart	22
Entity Relationship Diagram	23
Entity Relationship Diagram – Chen's Notation	23
Entity Relationship Diagram – Crow's Foot Notation	24
Hardware and Software Requirements	25

PART II METHODOLOGY

Project Title 26

Project Organization 26

Project Methodology 28

PART III SYSTEM TESTING AND ACCEPTANCE

Testing Methods and Approach 29

Test Cases 30

Maintenance Plan 33

Appendices

A. User Interface Design 34

B. User Manual 43

LIST OF TABLES

Table 1.0 Use Case Glossary 17

Table 3.0 Test Cases 30

Table 3.1 Maintenance Plan 33

LIST OF FIGURES

Figure 1.0 Activity Diagram for Account Authentication 3

Figure 1.1 Activity Diagram for Inventory Management 4

Figure 1.2 Activity Diagram for Sales Transaction 5

Figure 1.3 Activity Diagram for Daily Report 6

Figure 1.4 Class Diagram of the System 7

Figure 1.5 Context Level Data Flow Diagram of the System 8

Figure 1.6 Level 0 Data Flow Diagram of the System 9

Figure 1.7 Use Case Diagram 21

Figure 1.8 Gantt Chart 22

Figure 1.9 Entity-Relationship Diagram: Chen's Notation 23

Figure 1.10 Entity-Relationship Diagram: Crow's Foot Notation 24

Figure 2.0 Organizational Structure 26

Figure 2.1 Agile Methodology Model 28

Figure 3.0 Black-Box Testing Model 29

MENU IN THE USER SIDE

Figure 3.4 Product Management 36

Figure 3.5 Stock Management 36

Figure 3.6 Transaction Management 37

MENU IN THE ADMIN SIDE

Figure 3.7 Product Management 38

Figure 3.8 Stock Management 39

Figure 3.9 Transaction Management 39

Figure 3.10 Daily Report Management 40

Figure 3.11 Sales Report 41

Figure 3.12 Stock Report 41

Figure 3.13 User Management 42

PART I

Introduction

Introduction

The rapid advancement of information technology has significantly transformed business operations across various industries, including the healthcare sector. Pharmacies, in particular, occupy a critical role in ensuring the safe, accurate, and timely distribution of medicines to the public. Despite this important responsibility, many small to medium-sized pharmacies continue to depend on manual or semi-manual methods for managing their inventory and processing transactions. These conventional practices are consistently prone to human error, operational inefficiencies, and data inconsistencies, which can ultimately lead to critical issues such as stock shortages, the dispensing of expired medications, and inaccurate sales records.

An effective Pharmacy Inventory and Sales Management System is therefore essential to directly address these challenges by fully automating daily pharmacy operations. Such a system enables real-time monitoring of medicine stock levels, accurate tracking of sales transactions, and proper recording of product purchases and expiration dates. By seamlessly integrating inventory control with transaction processing, pharmacies can ensure that medicines are consistently well-stocked, expiration dates are closely monitored, and all financial records are maintained with precision. This integration significantly improves overall operational efficiency while simultaneously enhancing patient safety by substantially reducing the risk of dispensing expired or incorrect medications.

The development of PharmaTrack: Pharmacy Inventory and Sales Management System aims to deliver a reliable, user-friendly, and highly efficient solution for pharmacy staff. Through automation and centralized data management, the system is designed to support better decision-making, reduce staff workload, and minimize the errors commonly associated with manual record keeping. Ultimately, PharmaTrack contributes to improved service quality, stronger inventory control, and greater overall productivity within pharmacy operations.

General Objective

The general objective of this study is to design and develop a Pharmacy Inventory and Sales Management System that efficiently manages medicine inventory, sales transactions, and record-keeping processes to improve accuracy, efficiency, and reliability in pharmacy operations.

Specific Objectives

The following specific objectives guided the development of PharmaTrack:

- To develop a system that automates the recording and monitoring of medicine inventory, including stock levels and expiration dates.
- To implement a transaction management module that accurately records sales and generates real-time transaction reports.
- To provide an organized and user-friendly interface that enables pharmacy staff to efficiently manage inventory, process transactions, and generate reports for decision-making.

Data Flow Diagrams

The following diagrams illustrate how users and pharmacy administrators interact with the system during inventory management, product and stock updates, and the generation of daily sales and transaction reports.

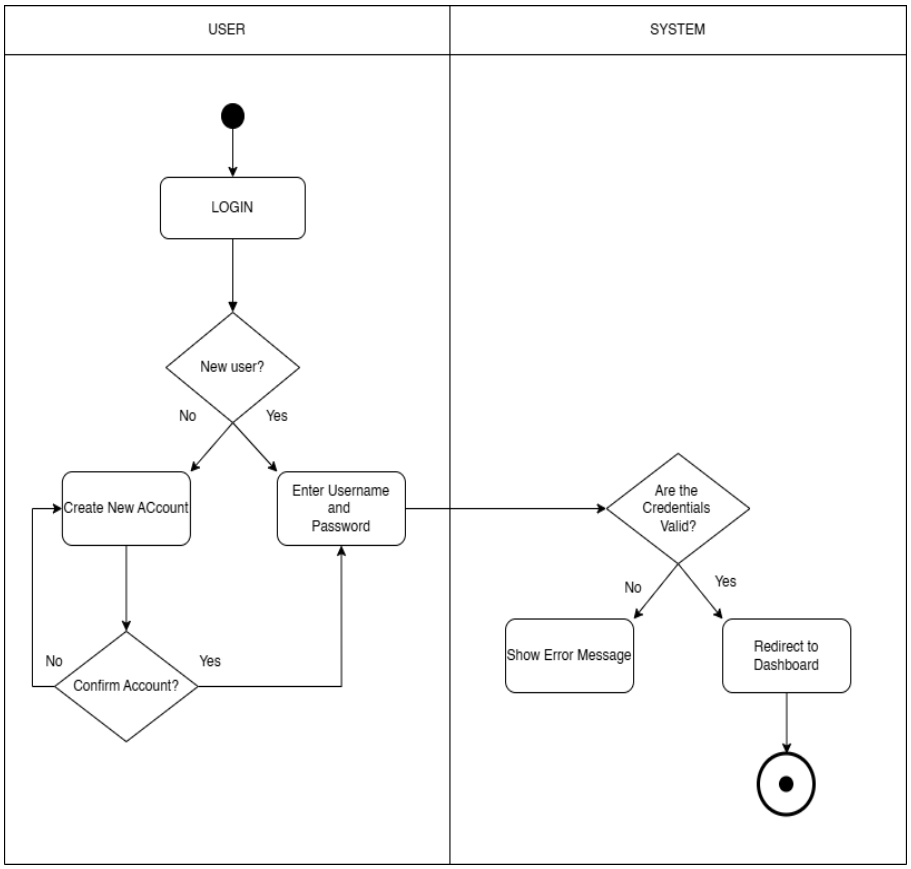


Figure 1.0 **<u>Activity Diagram for Account Authentication</u>**

Figure 1.0 illustrates the activity diagram depicting how each user or administrator registers and logs into the system. This process is fundamental to ensuring that only authorized personnel can access and manage inventory data, record stock level changes, and generate daily reports.

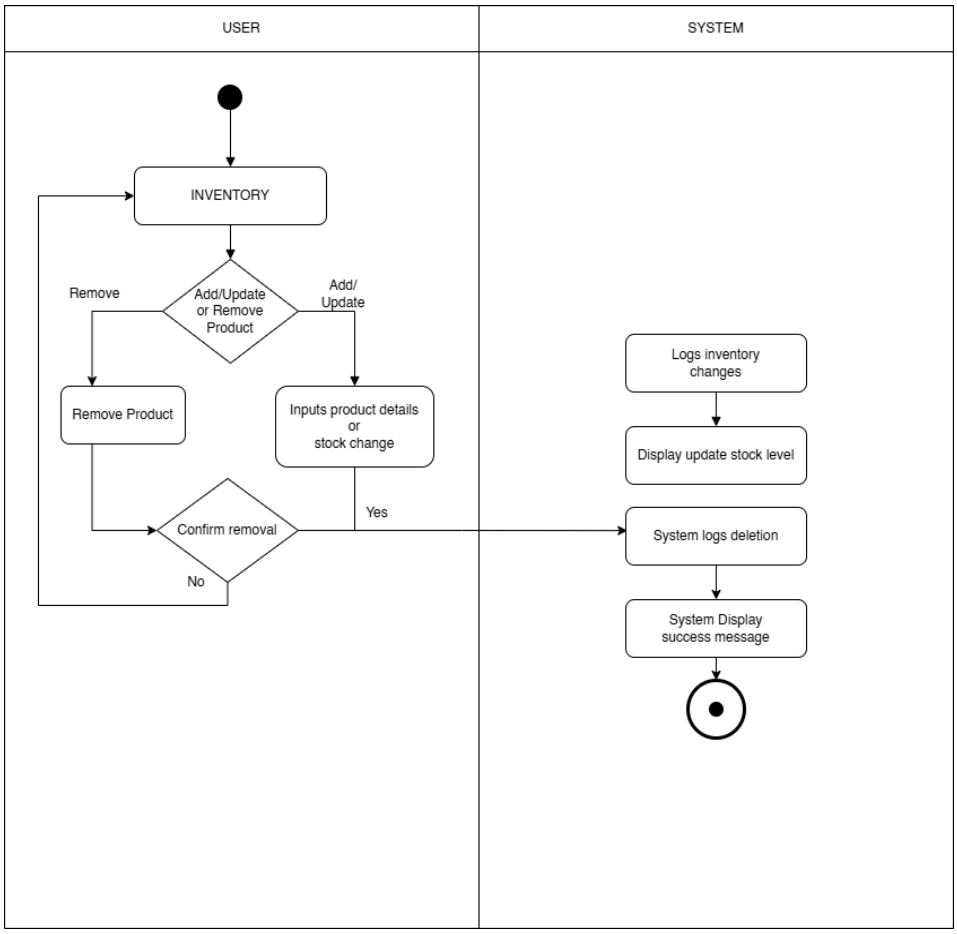


Figure 1.*1* **<u>Activity Diagram for Inventory Management</u>**

Figure 1.1 illustrates how the system manages inventory operations. Starting from the user accessing the inventory module, the user may add, update, or remove products, and the system validates and updates the database accordingly. The diagram also highlights how the system continuously monitors stock levels and records all changes to ensure accurate reporting.

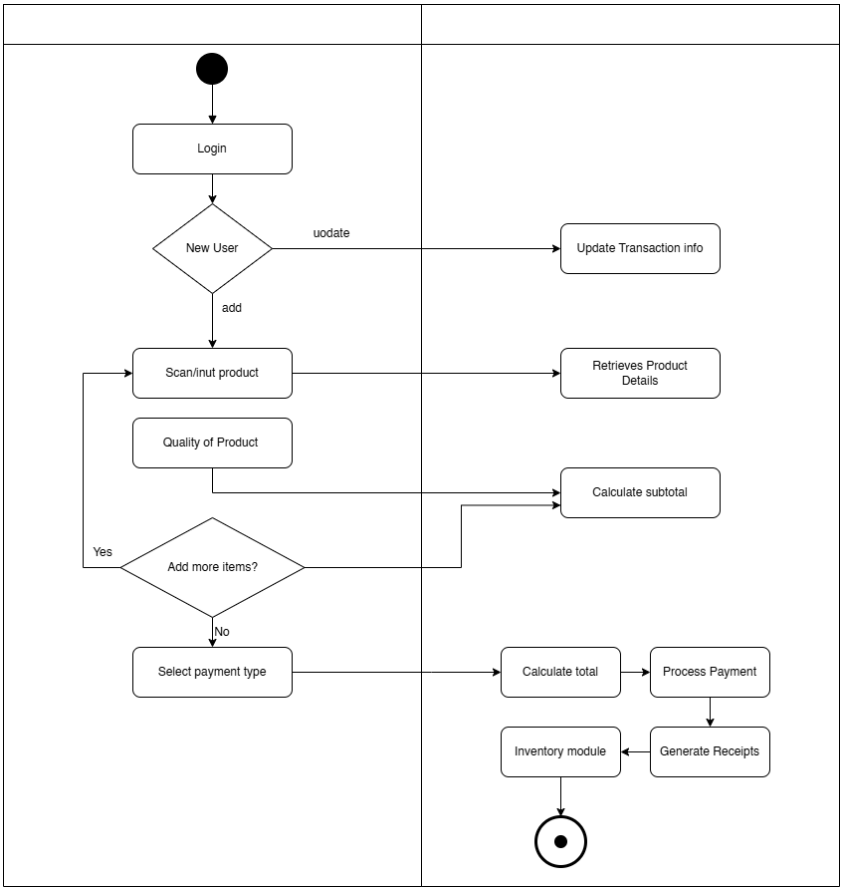


Figure 1.2 **Activity Diagram for Sales Transaction**

Figure 1.2 depicts the complete process of handling a sales transaction, beginning with product selection and quantity entry. The system retrieves product details, calculates the total amount, processes the payment, and updates the inventory by adjusting stock levels. The transaction is then permanently recorded, and a corresponding receipt is generated for reference.

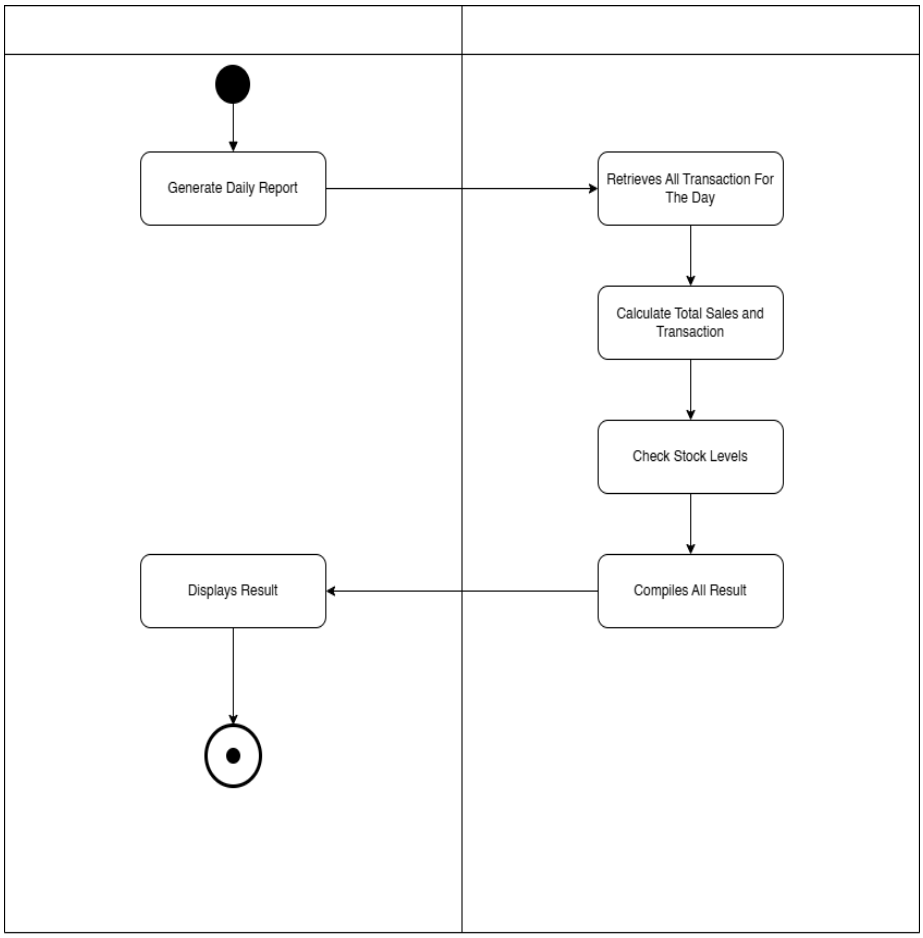


Figure 1.3 **Activity Diagram for Daily Report**

Figure 1.3 illustrates how the system generates daily reports by collecting transaction and inventory data accumulated throughout the day. The system processes and summarizes the information, then produces a comprehensive report displaying sales figures, stock changes, and other relevant operational metrics to support monitoring and informed decision-mak

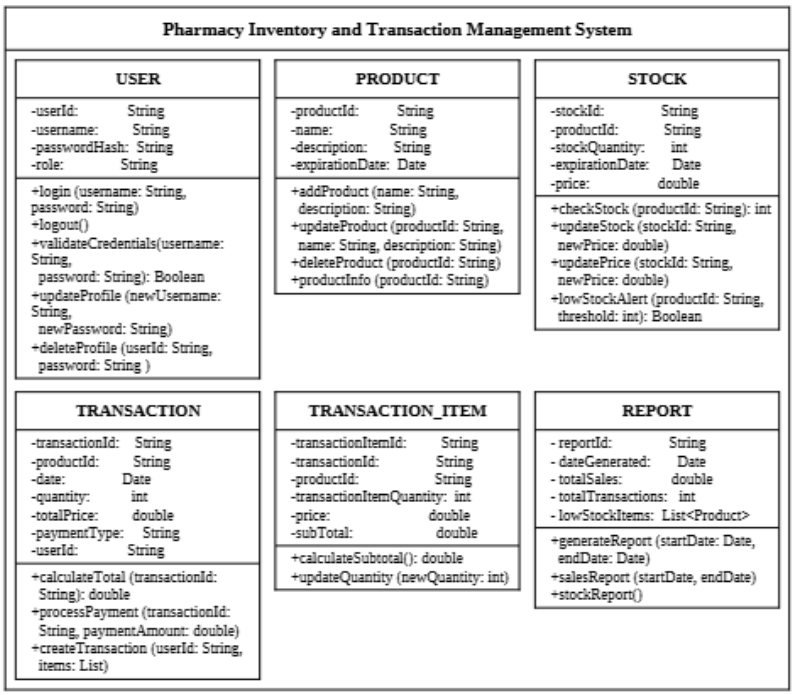


Figure 1.4 **Class Diagram of the System**

Figure 1.4 presents the class diagram of PharmaTrack, illustrating the system's structural design. It shows the key classes — User, Product, Stock, Transaction, Transaction Item, and Report — along with their attributes, methods, and the relationships that connect them to facilitate the system's core functionalities.

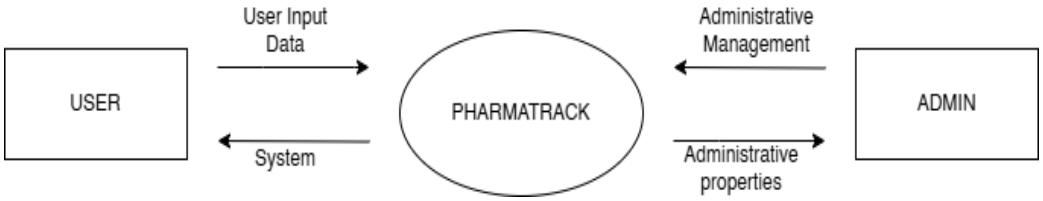
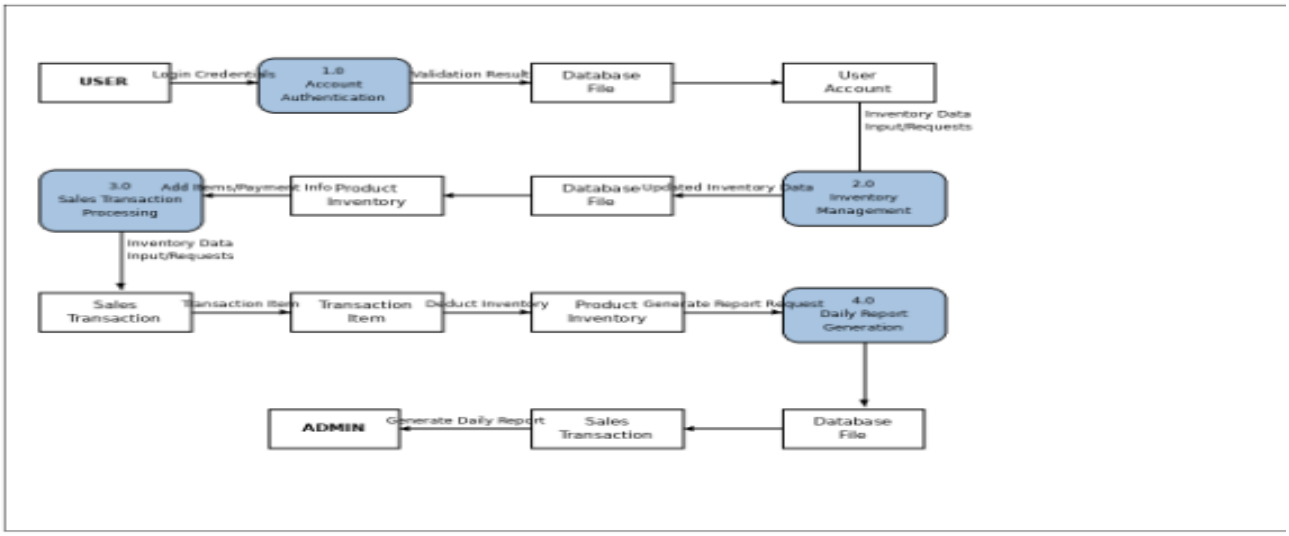


Figure 1.5 **Context Level Data Flow Diagram of the System**

Figure 1.5 presents the context-level data flow diagram, providing a high-level overview of the system's data interactions. It illustrates how external entities — specifically the User and the Admin — interact with PharmaTrack by submitting inputs such as product details, sales transactions, and inventory updates, and receiving outputs such as updated records and generated reports.

Figure 1.6 Level 0 Data Flow Diagram of the System



<u>Figure 1.6 Level 0 Data Flow Diagram of the System</u>

Figure 1.6 provides a more detailed decomposition of the system's major processes, including account authentication, inventory management, sales transaction processing, and daily report generation. It clearly illustrates how data flows between these processes, the data stores, and the system's external entities, and how raw inputs are transformed into meaningful and actionable outputs.

Functional Requirements

Based on the identified problems and operational needs of pharmacy businesses, the primary challenge stems from inadequate data management, causing errors in inventory tracking, order processing, sales recording, and reporting. These issues arise mainly from reliance on manual workflows, the absence of centralized information systems, and inconsistent data entry practices. The appropriate solution is to implement an integrated system that automates data handling, tracks inventory in real time, processes sales transactions accurately, and generates timely operational reports. To fully support both management and pharmacy staff, the system must also incorporate security controls, automated stock alerts, and comprehensive service tracking capabilities. The following functional requirements were identified based on careful observation and thorough analysis of typical pharmacy operations:

Account Authentication Module

I. Register User

a. Ability to register new users into the system.

b. All fields must be completely filled out before submission.

c. The following data fields are required:

i. UserID

ii. Username

iii. Password

iv. Role

II. Login User

a. Ability for users to log in to the system.

b. The system must verify user credentials before granting access.

c. The following data fields are required:

i. Username

ii. Password

III. Update User Details

a. Ability to update existing user account information.

b. The following data fields are required:

i. UserID

ii. Username

iii. Role

IV. Delete User

a. Ability to permanently delete a user account from the system.

b. The following data field is required:

i. UserID

V. Logout User

a. Ability for users to securely log out of the system.

VI. Read User Information

a. Ability to view the details of existing user accounts.

b. The following data fields are required:

i. UserID

ii. Username

iii. Role

Product Management

I. Add Product

a. Ability to add new products to the system.

b. All required fields must be completely filled out before submission.

c. The following data fields are required:

i. ProductID

ii. ProductName

iii. ProductPrice

iv. ProductQuantity

v. ProductExpiryDate

II. Read Product Information

a. Ability to view product details and current availability.

b. The following data fields are required:

i. ProductID

ii. ProductName

iii. ProductPrice

iv. ProductQuantity

v. ProductExpiryDate

III. Update Product Details

a. Ability to update existing product details in the system.

b. The following data fields are required:

i. ProductID

ii. ProductPrice

iii. ProductQuantity

iv. ProductName

v. ProductExpiryDate

IV. Delete Product

a. Ability to permanently delete a product record from the system.

b. The following data field is required:

i. ProductID

Stock Management

I. Stock In

a. Ability to add stock records for products.

b. The following data fields are required:

i. ProductStockID

ii. ProductID

iii. QuantityAdded

iv. DateOfStockIn

II. Read Stock Status

a. Ability to view a product's current stock levels.

b. The following data fields are required:

i. ProductStockID

ii. ProductID

iii. CurrentStockQuantity

III. Update Stock Record

a. Ability to update stock records whenever products are sold or restocked.

b. The following data fields are required:

i. ProductStockID

ii. ProductID

iii. QuantityAdded

iv. QuantityRemoved

v. DateOfStockIn

vi. DateOfStockOut

IV. Delete Stock Record

a. Ability to remove an existing stock record from the system.

b. The following data field is required:

i. ProductStockID

Sales Transaction Module

I. Create Transaction

a. Ability to create and record new sales transactions.

b. The following data fields are required:

i. TransactionID

ii. ProductID

iii. QuantitySold

iv. TotalPrice

v. Date

II. Read Transaction Information

a. Ability to view the details of recorded transactions.

b. The following data fields are required:

i. TransactionID

ii. QuantitySold

iii. TotalPrice

iv. Date

v. ProductID

III. Update Transaction

a. Ability to update the details of an existing transaction.

b. The following data fields are required:

i. TransactionID

ii. ProductID

iii. QuantitySold

iv. TotalPrice

v. Date

vi. DateOfUpdate

Daily Report Module

I. Daily Report

a. Ability to generate a daily report that consolidates both sales and stock information.

b. The system shall retrieve sales data using the SalesReportID and retrieve inventory data using the StockReportID.

c. The following data fields are required:

i. DailyReportID

ii. SalesReportID

iii. StockReportID

iv. ReportDate

v. TotalDailySales

vi. TotalTransactions

vii. LowStockItems

II. Sales Report

a. Ability to generate a sales report based on completed transactions.

b. The system shall calculate and display the total number of transactions within the specified period.

c. The following data fields are required:

i. SalesReportID

ii. TransactionDate

iii. TotalSales

iv. TotalTransactions

III. Stock Report

a. Ability to generate a comprehensive stock inventory report.

b. The system shall identify products with low stock levels and clearly display available stock quantities for all products.

c. The following data fields are required:

i. StockReportID

ii. ProductID

iii. ProductName

iv. AvailableQuantity

v. ExpirationDate

vi. StockStatus

Use Cases

Table 1.0

Use Case Glossary

Use-Case Name	Use-Case Description	Participating Roles and Actors
Account Authentication Module:		
Register User	This use case describes the process of creating a new user account and registering it within the system.	User, Admin
Login User	This use case describes the process of validating user credentials to grant authorized access to the system.	User, Admin
Update User Details	This use case describes the process of modifying existing user account information.	Admin
Delete User	This use case describes the process of permanently removing a user account from the system.	Admin
Logout User	This use case describes the process of securely logging out of the system.	User, Admin
Read User Information	This use case describes the process of viewing the details of registered user accounts.	Admin
Product Management		
Add Product	This use case describes the process of adding new products to the system inventory.	User, Admin
Read Product Information	This use case describes the process of viewing product details and availability status.	User, Admin
Update Product Information	This use case describes the process of modifying existing product details in the system.	User, Admin
Delete Product	This use case describes the process of permanently removing a product record from the system.	Admin

Stock Management		
Stock In	This use case describes the process of adding stock quantities to the inventory for a specific product.	User, Admin
Read Stock Status	This use case describes the process of viewing the current stock levels of products in the inventory.	User, Admin
Update Stock Record	This use case describes the process of modifying existing stock records in the system.	User, Admin
Delete Stock Record	This use case describes the process of permanently removing a stock record from the system.	Admin
Sales Transaction Module		
Create Transaction	This use case describes the process of recording a new sales transaction in the system.	User, Admin
Read Transaction Information	This use case describes the process of viewing the details of recorded transactions.	User, Admin
Update Transaction	This use case describes the process of modifying an existing sales transaction record.	Admin
Daily Report Module		
Daily Report	This use case describes the process of generating a daily summary consolidating both sales and inventory information.	Admin
Sales Report	This use case describes the process of generating a sales report that summarizes all completed transactions for a specified period.	Admin
Stock Report	This use case describes the process of generating a stock report that displays the current inventory status of products, including quantities, expiration dates, and low-stock items.	Admin

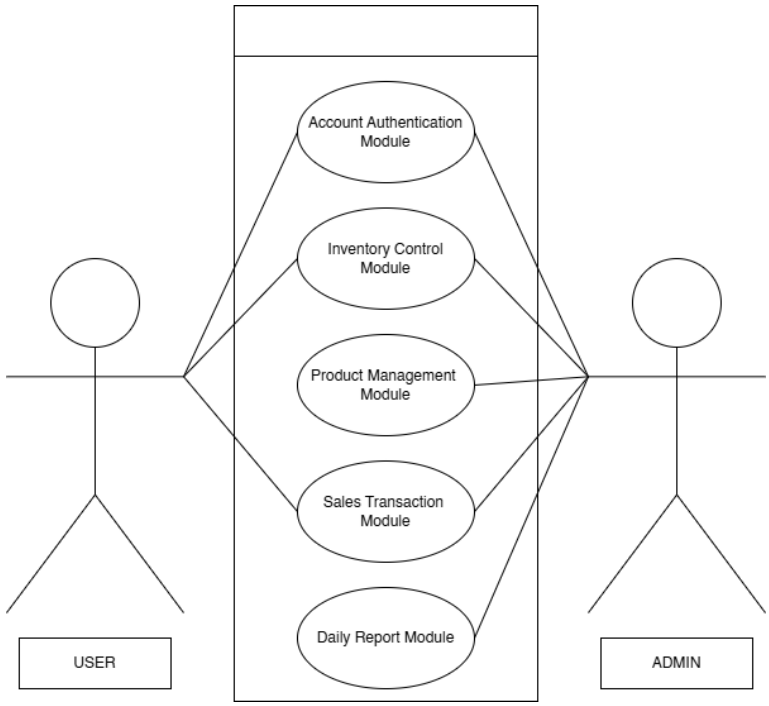


Figure 1.7 Use Case Diagram

Task Name	Duration	Start	End	November	December	January	February
SPRINT 1	47 days	11/24/25	01/31/26				
Planning: Requirements Gathering	7 days	11/24/25	12/07/25				
Design: System Architecture Design	14 days	12/08/25	12/14/25				
Development: Core Feature Development	7 days	01/06/26	01/19/26				
Testing: System Testing & Debugging	14 days	01/20/26	01/26/26				
Deployment: Deployment & Configuration	3 days	01/27/26	01/29/26				
Review: Final Evaluation & Revision	2 days	01/30/26	01/31/26				
SPRINT 2	23 days	02/01/26	02/23/26				
Planning: Backlog Creation	3 days	02/01/26	02/03/26				
Design: UX/UX Design & Prototyping	5 days	02/04/26	02/08/26				
Development: Database & Module	10 days	02/09/26	02/18/26				
Testing: User Acceptance Testing	3 days	02/19/26	02/21/26				
Review: Documentation & Submission	2 days	02/22/26	02/23/26				

Figure 1.8 Gantt Chart

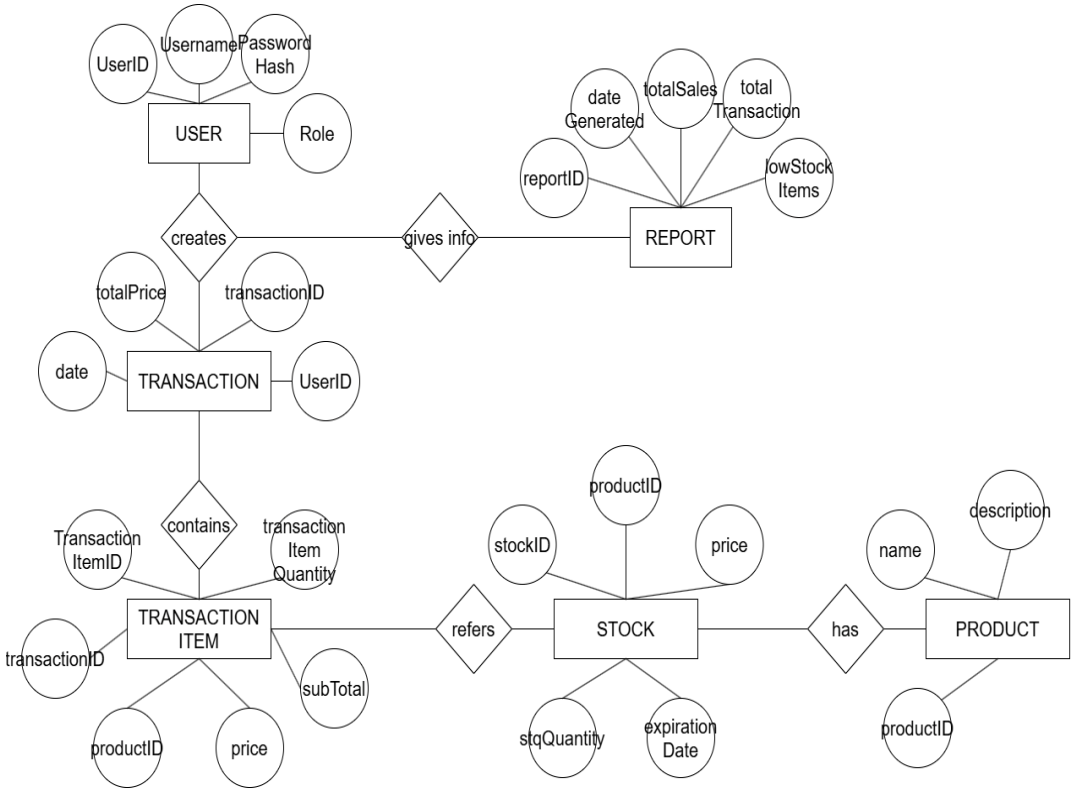


Figure 1.9 **<u>Entity-Relationship Diagram: Chen's Notation</u>**

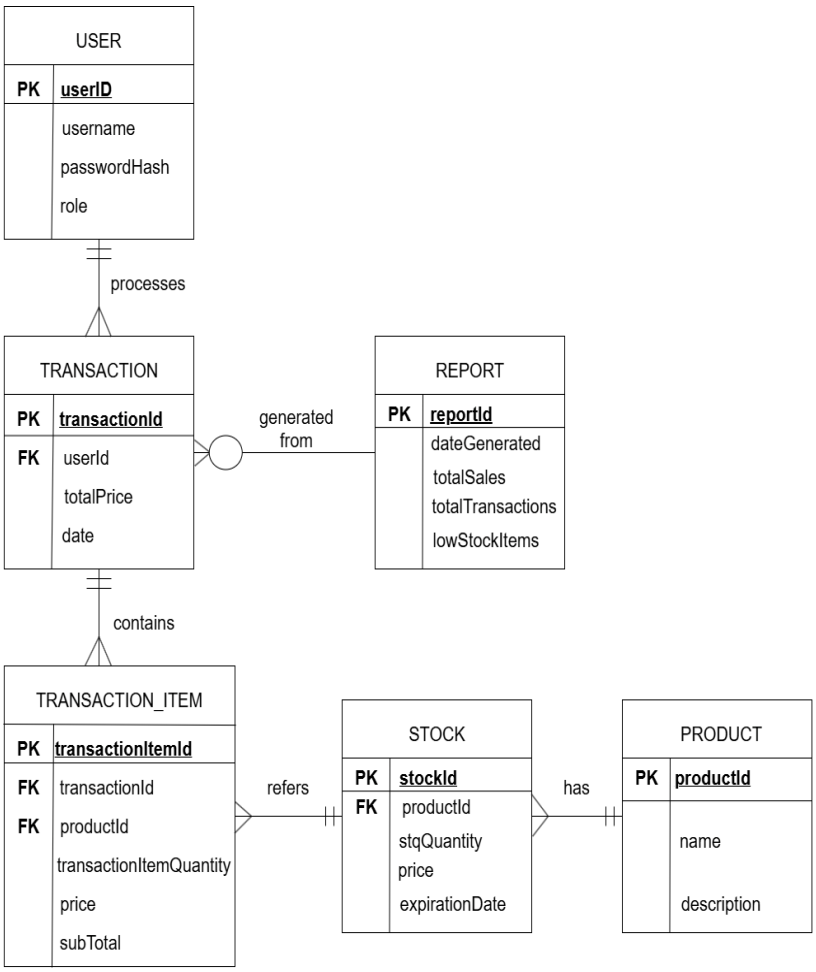


Figure 1.10 **Entity-Relationship Diagram: Crow's Foot Notation**

Hardware and Software Requirements

Minimum Software Requirements

Operating System: Windows 10 (64-bit) or higher

Minimum Hardware Requirements

Processor: Intel Core i3 (4th Generation) or higher

RAM: Recommended 8 gigabytes

Hard Disk Space: 128 gigabytes HDD/SSD

Graphics Card: Integrated Graphics (Intel HD / AMD Vega)

Display: Recommended resolution of 1920 x 1080

PART II

METHODOLOGY

Project Title

The project title proposed by the developers is PharmaTrack: Pharmacy Inventory and Sales Management System. The developers have selected this title because it directly reflects the main functionalities of the proposed system for pharmacy business operations. It highlights the system's capacity to efficiently manage inventory, monitor stock levels, and process sales transactions accurately. This title clearly complements the operational needs of the business by emphasizing tracking, management, and overall system efficiency.

Project Organization

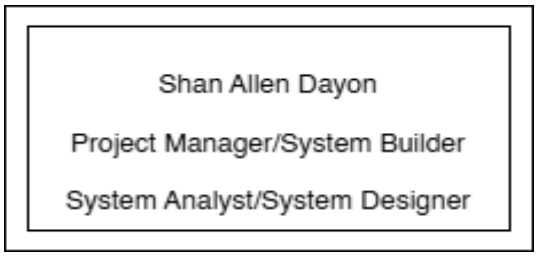


Figure 2.0 **Organizational Structure**

Project Manager – The Project Manager is the individual who leads the project from the planning phase through to its full execution. This role entails organizing tasks, managing team members, allocating resources, and ensuring that the project's objectives are achieved within the specified timeline and scope. The Project Manager holds full responsibility and authority over the successful completion of the system.

System Analyst – The System Analyst is responsible for analyzing business requirements and identifying the functional and technical needs of the system. This role interprets organizational problems and translates them into viable system solutions. The System Analyst prepares the Requirements Specification document and ensures that the system is designed to meet all defined user needs.

System Designer – The System Designer develops the overall system architecture and design plan. This includes creating the database structure, system diagrams, and interface layouts. The primary output of this role is the Design Specification, which serves as the comprehensive blueprint for the system development process.

System Builder – The System Builder is responsible for coding and constructing the system based on the approved design specifications. This role ensures that all modules are properly implemented, integrated, and fully functional. The System Builder plays a central role in transforming system designs into a working and operational application.

Project Methodology

Figure 2.1 Agile **Methodology Model**

The development of PharmaTrack: Pharmacy Inventory and Sales Management System follows the Agile Software Development Methodology. Agile focuses on iterative development, continuous feedback, and flexibility throughout the entire project lifecycle. Rather than completing the entire system in a single prolonged process, the project is divided into manageable and incremental phases that are repeated in cycles to continuously improve the system. The process begins with the Requirements phase, where the system's functional and non-functional needs are identified based on pharmacy operations. This is followed by the Design phase, in which system architecture, database structure, and process diagrams are created. During the Implementation phase, the system modules are developed using Java and MySQL. The Testing phase ensures that each feature functions correctly and meets all user requirements. After testing, the system proceeds to the Deployment phase, where it is installed and prepared for actual operational use. Finally, a Review phase is conducted to evaluate overall system performance, gather user feedback, and identify areas for improvement in the next deployment cycle.

PART III

SYSTEM TESTING AND ACCEPTANCE

Testing Methods and Approach

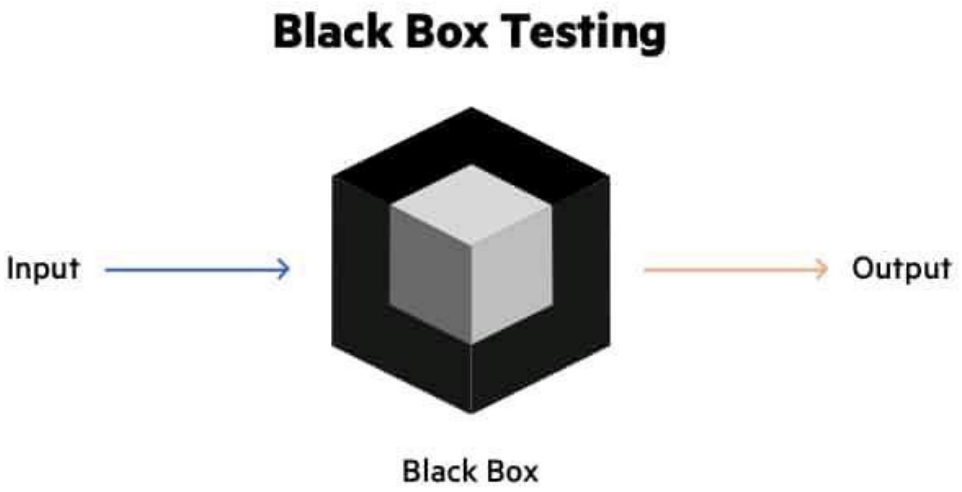


Figure 3.0 **Black-Box Testing Model**

Black-Box Testing

PharmaTrack: Pharmacy Inventory and Sales Management System will undergo comprehensive system testing to ensure that all features function correctly and consistently meet the specified requirements. The testing process will focus on validating the system's performance, reliability, usability, and accuracy in handling all pharmacy operations.

The system will employ Black Box Testing, which evaluates the software's functionality without requiring any knowledge of its internal code or logical structure. Test cases will be prepared for all major modules, including user login, product management, stock monitoring, sales transaction processing, and report generation. Each function will

be tested to verify correct input processing, accurate output generation, and proper error handling.

To ensure realistic evaluation, selected users will interact with the system under normal operating conditions. Following their testing sessions, users will complete an evaluation questionnaire to assess the system's usability, efficiency, and overall performance. The collected data will be thoroughly analyzed to identify any remaining issues and recommend necessary improvements prior to final deployment.

Test Cases

Table 3.0

Test Cases

Test Case ID	Test Case Description	Input	Expected Output
AA-01	Login with valid credentials	Correct username and password	User is successfully logged in and directed to the dashboard
AA-02	Login with incorrect password	Valid username, incorrect password	Error message is displayed indicating invalid credentials
AA-03	Login with unregistered account	Unregistered username	Login is denied and an error message is displayed
AA-04	Login with empty fields	Blank username and password	Validation message is shown prompting the user to fill in all fields
AA-05	Logout from the system	Click the logout button	User is successfully logged out and redirected to the login screen
AA-06	Update user account details	Valid updated account information	Account details are updated successfully in the database
AA-07	Delete user account	Admin confirms account deletion	User account is successfully removed from the system
AA-08	Login using a deleted account	Credentials of deleted account	Login is denied and an appropriate error message is displayed
IC-01	Add a new medicine	Complete medicine details	Medicine is successfully added to the inventory
IC-02	Add medicine with missing data	Incomplete medicine details	Validation error is shown; medicine is not added
IC-03	Update medicine information	Modified medicine data	Medicine details are successfully updated in the database
IC-04	Delete medicine record	Select medicine and confirm deletion	Medicine record is permanently removed from the inventory
IC-05	Search for a medicine	Medicine name or keyword	Correct medicine record is retrieved and displayed

IC-06	Add stock to an existing medicine	Valid restock quantity	Stock quantity is correctly increased in the inventory
IC-07	Enter invalid stock quantity	Negative or zero quantity	Validation error is displayed; stock is not updated
IC-08	Low-stock detection	Stock reaches the pre-defined low threshold	Low-stock warning alert is displayed to the user
IC-09	No-stock detection	Stock quantity reaches zero	Out-of-Stock status is displayed for the product
IC-10	Expired medicine detection	Expired medicine record present in the system	Expiration alert notification is shown
ST-01	Process a sale with sufficient stock	Valid medicine and quantity	Sale transaction is completed and recorded successfully
ST-02	Process a sale with low stock	Quantity equals remaining available stock	Sale completes with a low-stock warning displayed
ST-03	Process a sale with no stock	Select an out-of-stock medicine	Transaction is denied with an appropriate error message
ST-04	Process a sale with insufficient stock	Quantity exceeds available stock	Transaction is denied with an error message
ST-05	Calculate total transaction amount	Multiple medicines selected	Correct total amount is calculated and displayed
ST-06	Complete a sale transaction	Click confirm sale button	Transaction is successfully recorded in the database
ST-07	Cancel a sale transaction	Click the cancel button	Transaction is discarded and the form is closed
ST-08	Deduct stock after a sale	Completed sale transaction	Inventory is correctly updated by deducting the sold quantity
ST-09	Invalid quantity input	Non-numeric or negative value entered	Validation error is displayed and transaction is blocked
ST-10	Update transaction quantity	Modify quantity of sold item	Transaction record is updated successfully
ST-11	Update transaction to higher/lower quantity	Increase or reduce quantity of sold item	Transaction and stock levels are adjusted accordingly

ST-12	Update transaction for a low-stock item	Modify a sale with limited remaining stock	Warning is displayed; update proceeds if quantity is valid
ST-13	Update transaction for an out-of-stock item	Modify a sale when stock is zero	Update is denied with an appropriate error message
DR-01	Generate daily sales report	Select the current date	Daily sales report is successfully generated and displayed
DR-02	Generate report with no sales	Select a date with no recorded transactions	A zero sales report is generated for the selected date
DR-03	View daily report details	Open the generated report	Correct and complete sales data is displayed
DR-04	Verify low-stock items in report	Generate the stock report	All low-stock items are correctly listed in the report
DR-05	Verify out-of-stock items in report	Generate the stock report	All out-of-stock items are correctly identified and listed

Maintenance Plan

Table 3.1

Maintenance Plan

Maintenance Item	Daily	Weekly	Monthly	6 Months	Yearly
System monitoring and log review	✓				
Backup of system and database	✓				
Routine debugging		✓			
Major updates and system enhancements			✓		
User feedback evaluation				✓	
security audit				✓	

USER INTERFACE DESIGN

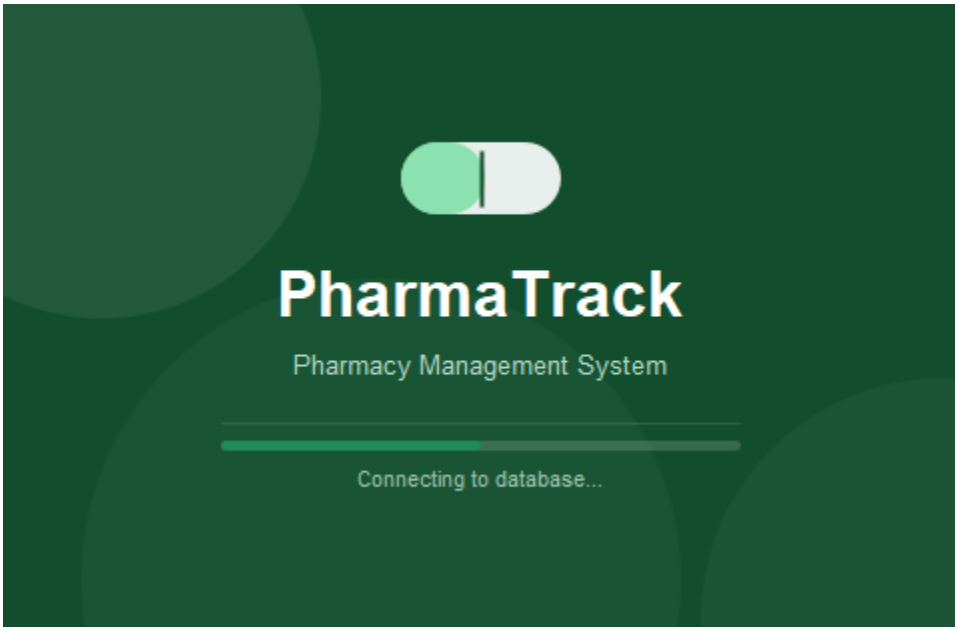


Figure 3.1 **Splash Screen**

This figure shows the logo and at the same time, the system title. This will load up every time the system is used

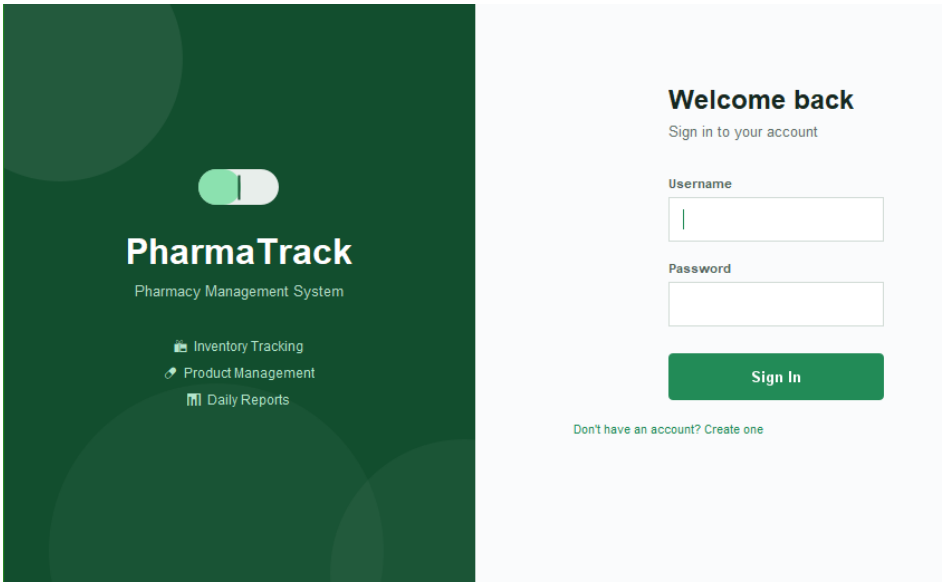


Figure 3.2 **Login Screen**

This figure shows the welcome login screen, which requires a valid username and password that should be registered in the system's database.

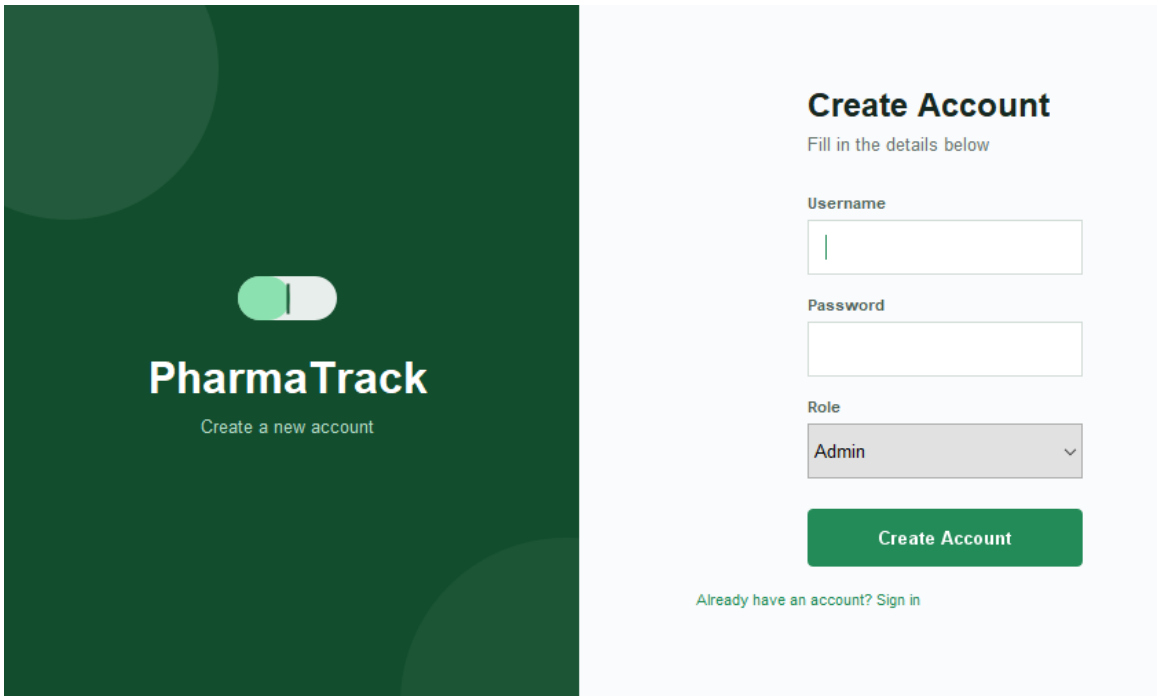


Figure 3.3 **Register Screen**

This figure shows the register screen, which requires a username, password, and the role for the user, and these will be registered in the system's database.

MENU IN THE USER SIDE

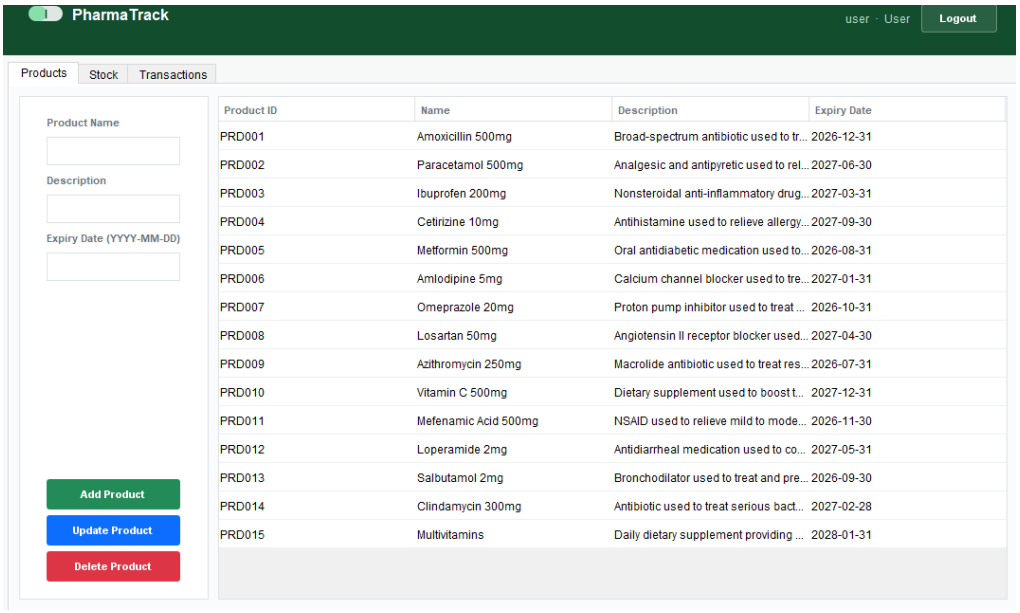


Figure 3.4 **Product Management**

In this form, the user can manage the product inventory. The user can add, update, and delete products. The user could also view certain product information. In the form, the Product ID, its name, description, and its expiry date could be seen.

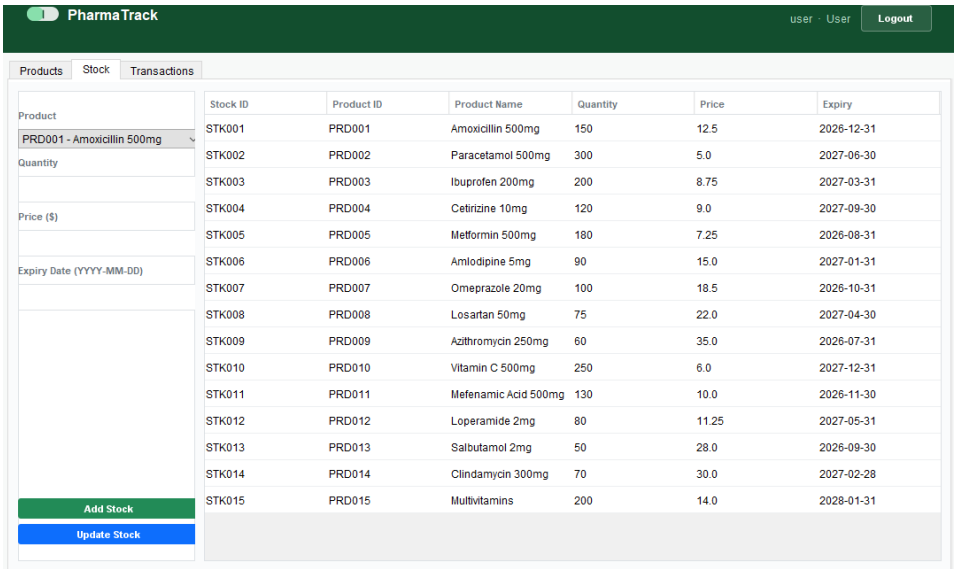
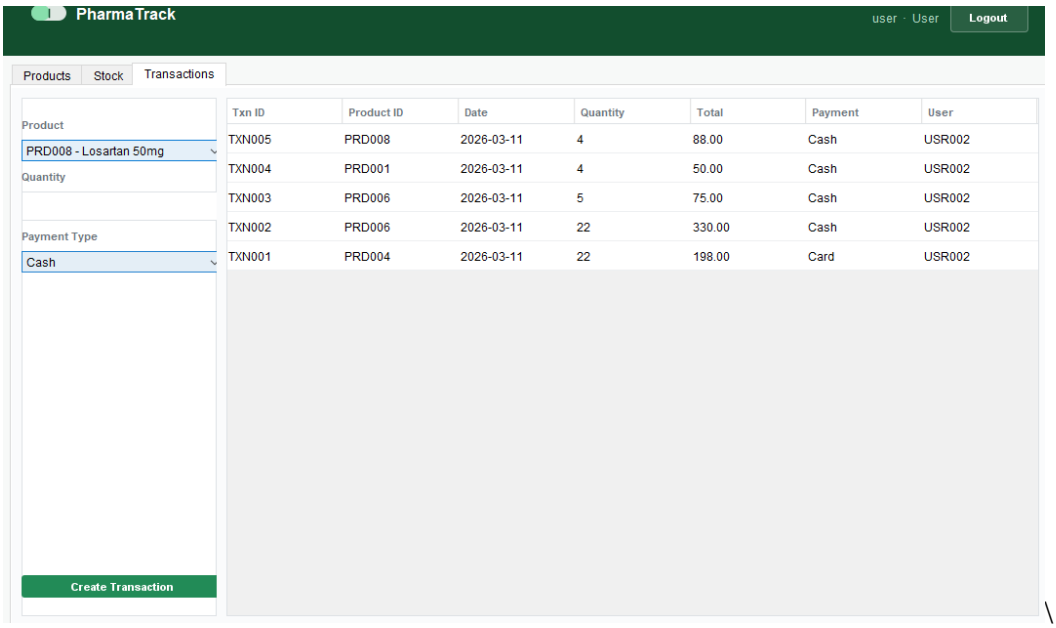


Figure 3.5 **<u>Stock Management</u>**

In this form, the user can manage the stock inventory. The user can add, view, and update the stocks. In the form, the Stock ID, Product ID, its name, quantity, price, and its expiry date could be seen.



\

Figure 3.6 **Transaction Management**

In this form, the user can manage the transactions of the products. The user can create a transaction. In the form, the Transaction ID, Product ID, the date, quantity, price, payment method and the User ID of who conducted the transaction could be seen.

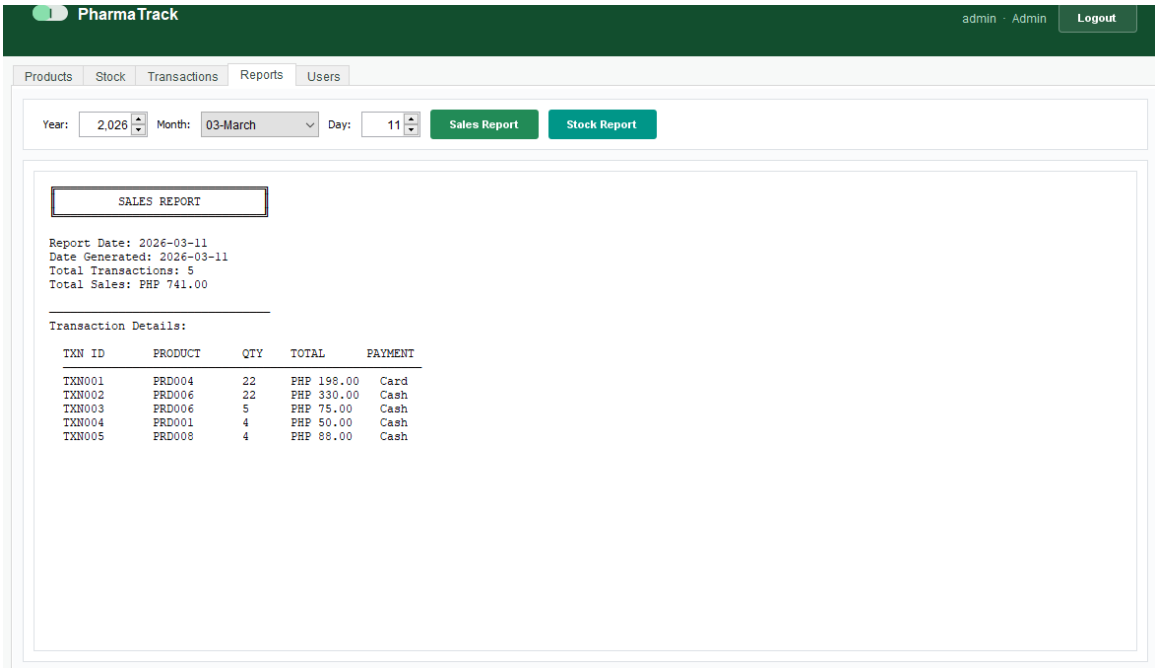


Figure 3.10 **Daily Report Management**

The admin handles the daily report management. In this tab, the Sales Report and Stock Report will be generated, which also depends on the date.

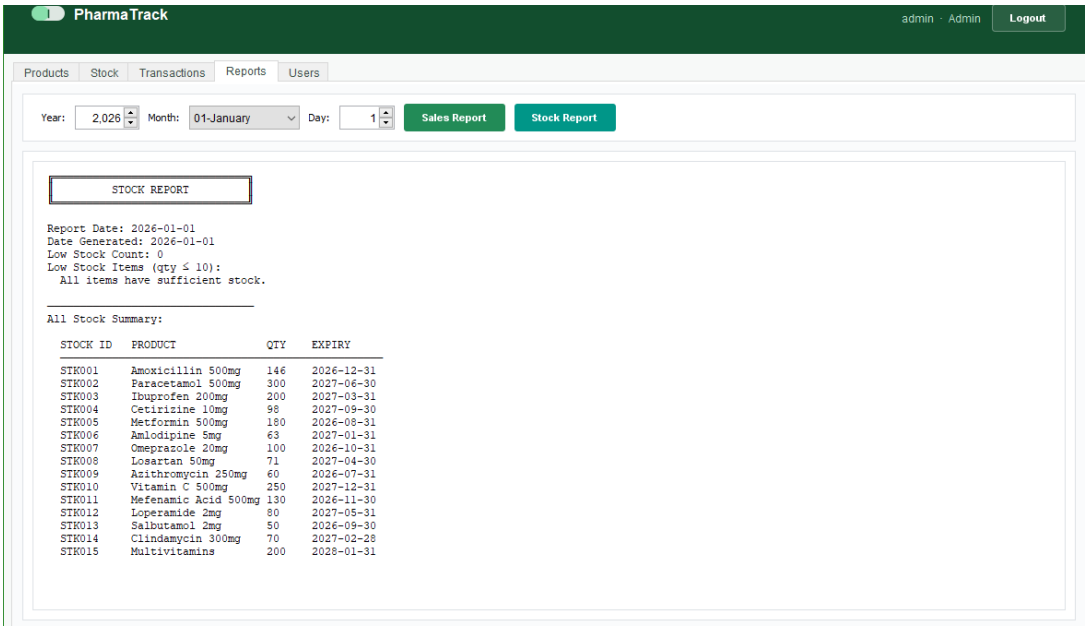


Figure 3.11 **Sales Report**

Once the Sales Report button is clicked by the admin, the Sales Report of a certain date will be generated. In the table, the report date, date generation, total transactions and total sales can be seen.

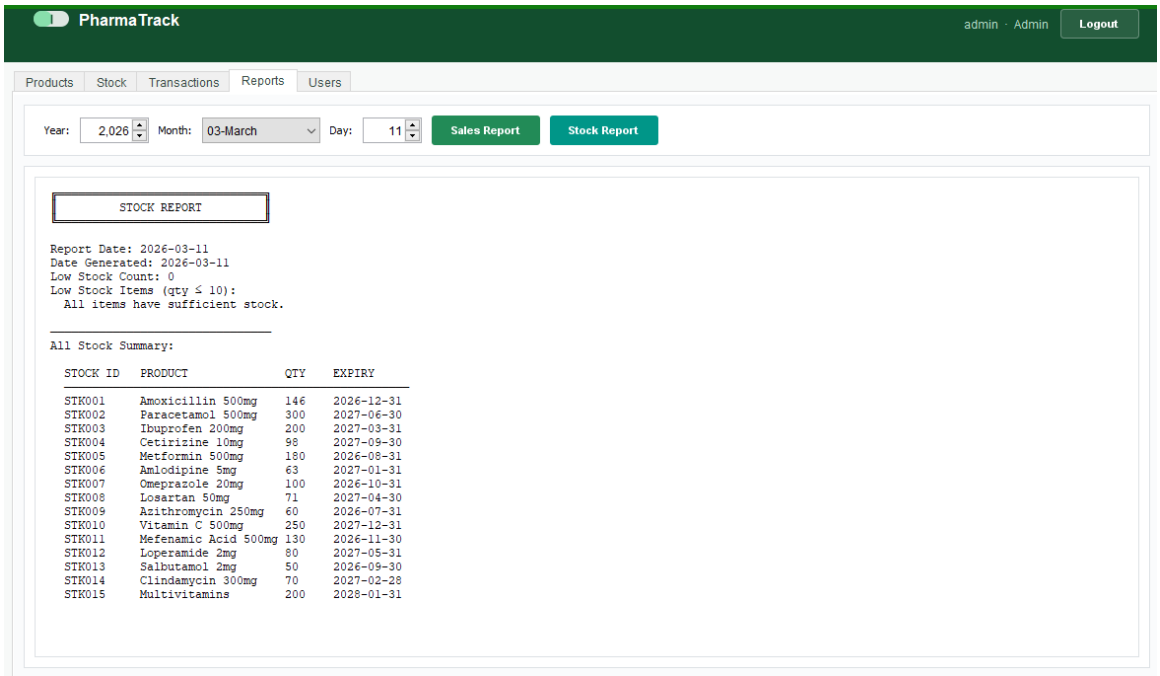


Figure 3.12 **Stock Report**

Once the Stock Report button is clicked by the admin, the Stock Report of a certain date will be generated. In the table, the report date, date generation, low stock count, and low stock items can be seen.

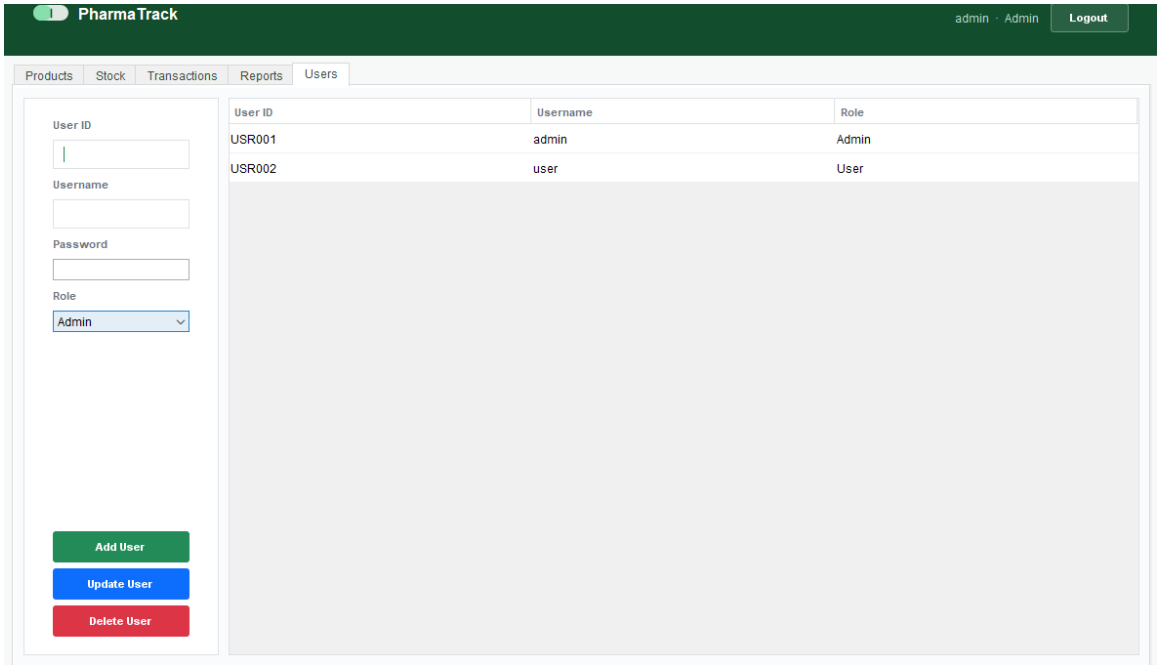


Figure 3.13 **User Management**

The admin handles the user management. Instead of registering, the admin can add a user, update the information of a user, and delete a user. This is done so that the admin can also manage on whoever's using the system. In the form, the User ID, username, and role can be seen.

USER MANUAL

Pharmatrack: Pharmacy Inventory and Transaction Management System

March 2026

TABLE OF CONTENTS

Title	Page Number
Section 1:	
Introduction	**Error! Bookmark not defined.**
System Unit Requirements	1
Section 2:	2
The Information System	2
a.) Splash Screen	2
b.) Login Screen	3
c.) Register Screen	3
Features for both Admins and Users: a.) Menu	4
i.) Product Management	4
ii.) Stock Management	5
iii.) Transaction Management	7
Features for Admins only: a.) Menu	9
i.) Daily Report Management	9
ii.) User Management	9
Section 3: Contact Us	11

SECTION 1

INTRODUCTION

Pharmacy Inventory and Transaction Management System aims to provide an automated inventory and sales transaction management solution. The project aims to lessen the workload of pharmacy staff in order for them to save time, effort, and improve operational efficiency. The project also aims to ensure accurate recording of medicine stocks and transactions, reducing errors and maintaining reliable records, especially since it involves critical healthcare services and financial transactions.

SYSTEM UNIT REQUIREMENTS

Basic Requirement:

HARDWARE

1.) A Desktop Unit or a Laptop Unit with at least Intel Core i3 processor or equivalent.
2.) At least 8 GB RAM.
3.) At least 10 GB HDD/SSD storage.
4.) Integrated Graphics (Intel HD/UHD or AMD Vega).
5.) Display with 1366 x 768 resolution or higher.
6.) Standard keyboard and mouse.
7.) Internet connection for database access and system updates.

SOFTWARE

1.) Operating System must be Microsoft Windows 10 or higher.
2.) Java Development Kit (JDK 8 or higher).

3.) MySQL for database management.

To download Java Development Kit 8, refer to the link below:
http://www.oracle.com/technetwork/java/javase/downloads/jdk8-downloads-2133151.html

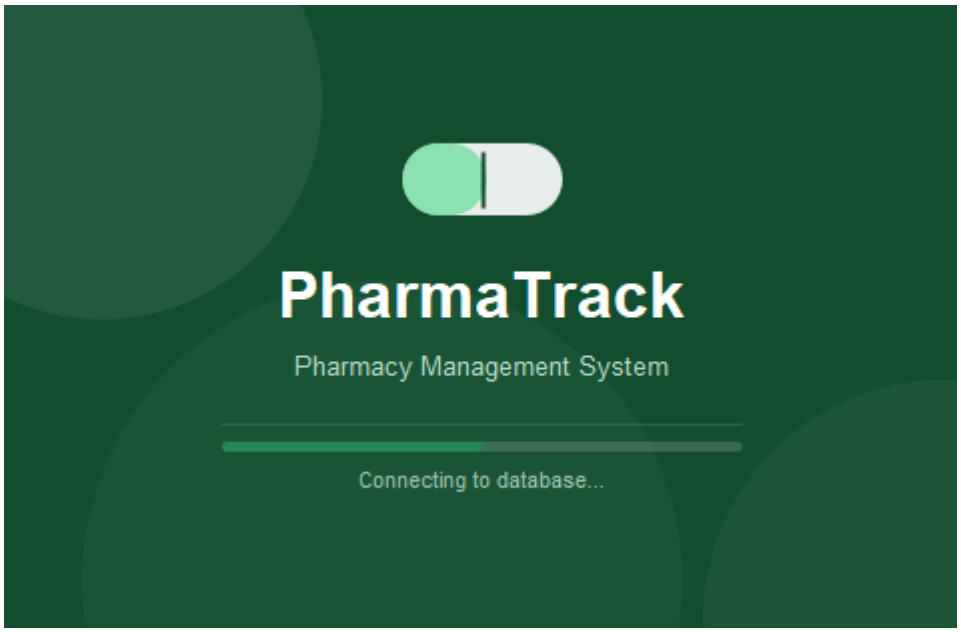
To download MySQL, refer to the link below:

https://dev.mysql.com/downloads/installer/

SECTION 2

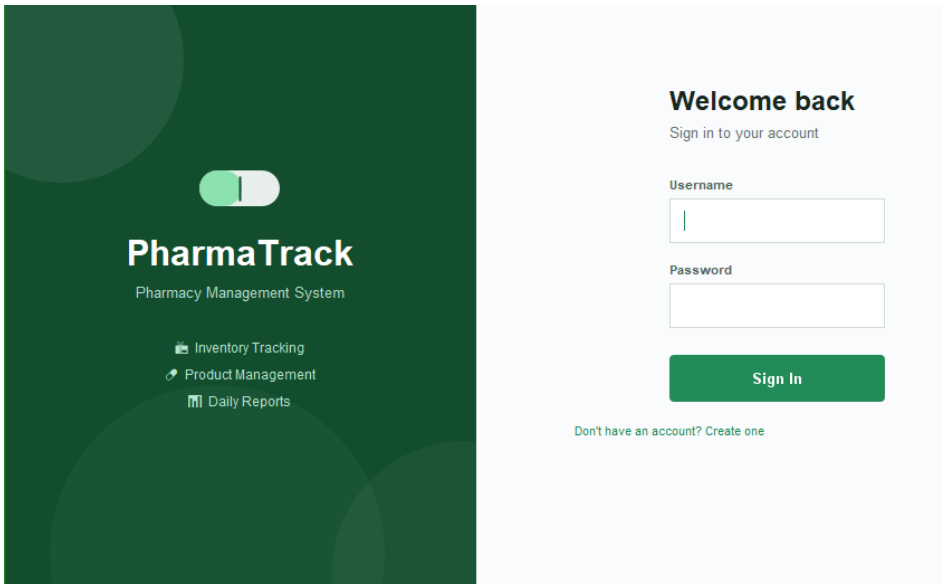
The Information System

a.) Splash Screen



Step 1: Wait for the loading screen to load

b.) Login Screen



Step 1: Login with a registered username and password

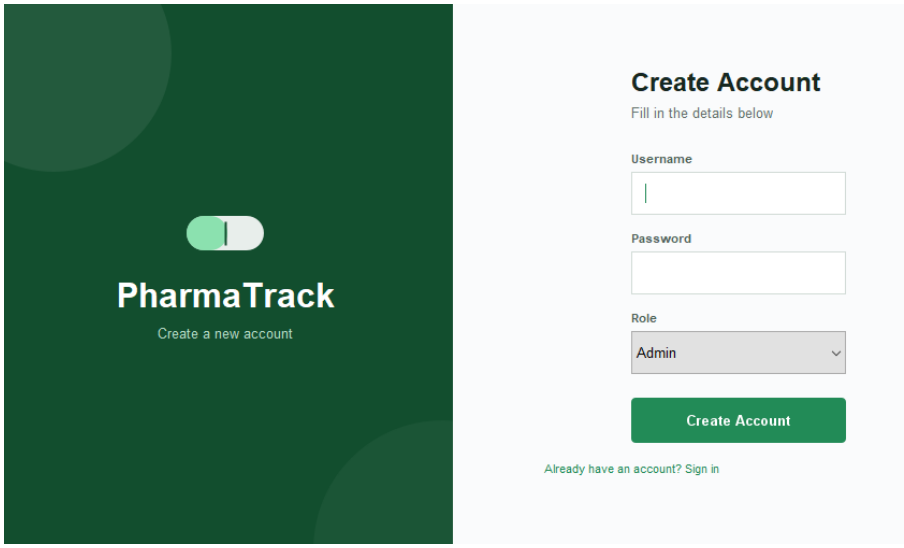
Step 2: Press the blue Login button

To exit the system, press X button or simply press Alt+f4.

To minimize the system, press the – button.

To go to the Register Screen, click the Register button

c.) Register Screen



If the user/admin doesn't have a registered account:

Step 1: Click the Register button in the Login screen

Step 2: Fill up the needed information

Step 3: Click the blue Register button

Step 4: Login using the newly registered account

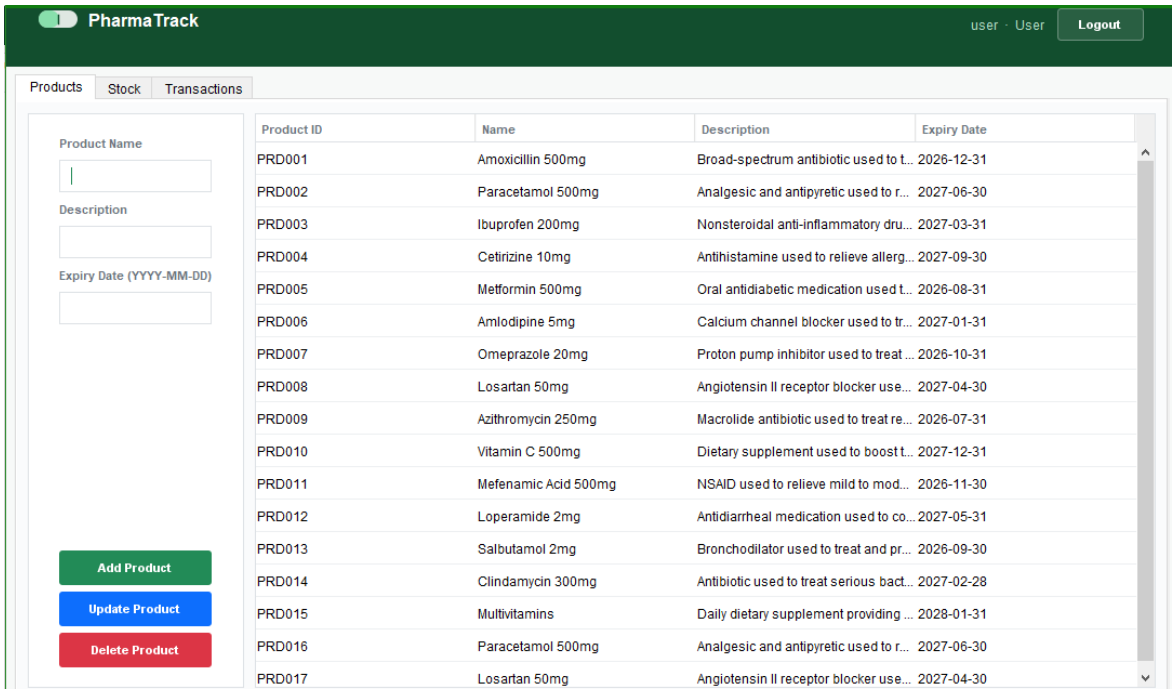
To exit the system, press X button or simply press Alt+f4.

To minimize the system, press the – button.

To return to Login Screen, click the Back to Login button

<u>Features for both Admins and Users:</u> **a.) Menu**

i.) Product Management



To add a product:

Step 1: Fill in the Product Name, Description, and Expiry Date

Step 2: Click the Add Product button

Step 3: Confirm the notification

To update a product:

Step 1: Click a row from the table, and it will display its information in the fields above

Step 2: Update the information

Step 3: Click the Update Product button

Step 4: Confirm the notification

To delete a product:

Step 1: Click a row from the table, and it will display its information in the fields above

Step 2: Click the Delete Product button

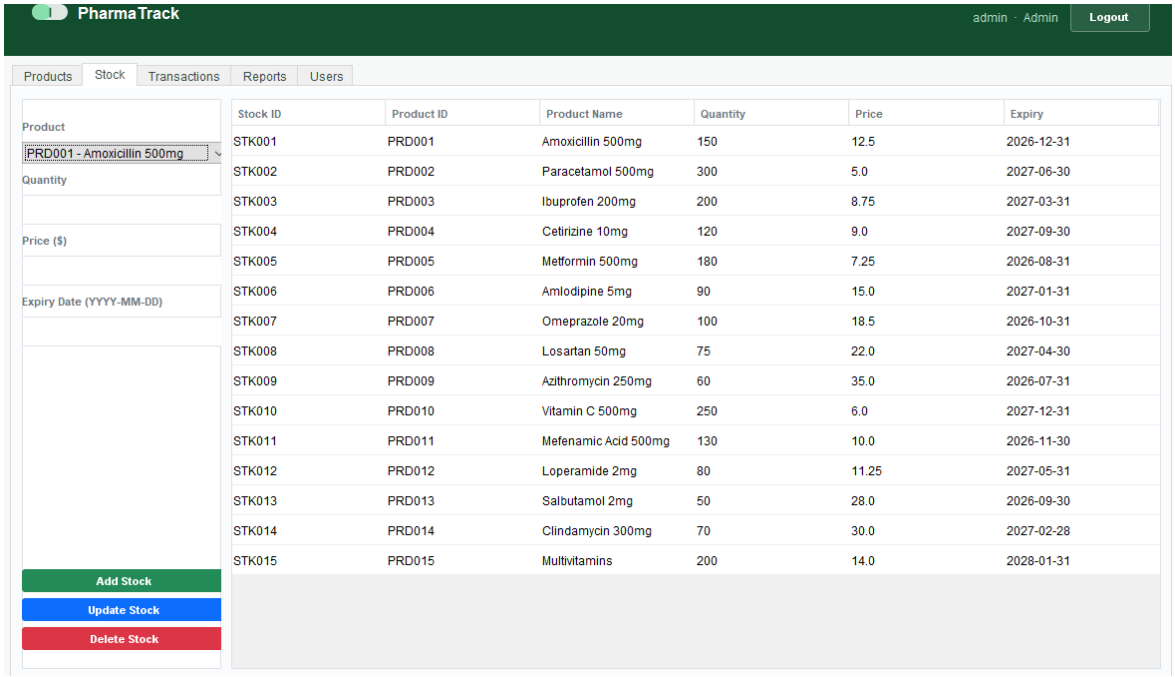
Step 3: Confirm the notification

To exit the system, press X button or simply press Alt+f4.

To minimize the system, press the – button.

To return to Login Screen, click the Logout button.

ii.) Stock Management



To add a stock:

Step 1: Click a row from the table or choose a product from the dropdown

Step 2: Fill in the Quantity, Price and its Expiry Date

Step 3: Click the Add Stock button

Step 4: Confirm the notification

To update a stock:

Step 1: Click a row from the table, and it will display its information in the fields above

Step 2: Update the information

Step 3: Click the Update Stock button

Step 4: Confirm the notification

FOR ADMINS

To delete a stock:

Step 1: Click a row from the table, and it will display its information in the fields above

Step 2: Click the Delete Product button

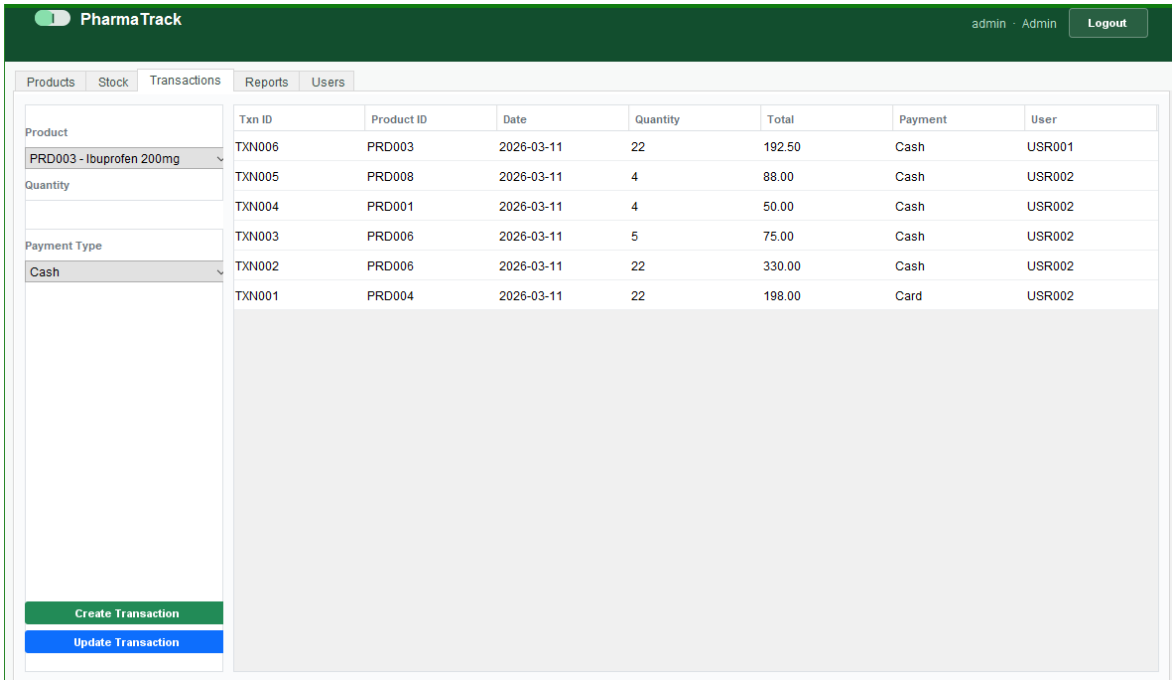
Step 3: Confirm the notification

To exit the system, press X button or simply press Alt+f4.

To minimize the system, press the – button.

To return to Login Screen, click the Logout button.

iii.) Transaction Management



To create a transaction:

Step 1: Click a row from the table or choose a product from the dropdown

Step 2: Fill in the Quantity and the Payment Type

Step 3: Click the Create Transaction button

Step 4: Confirm the notification

FOR ADMINS

To update a transaction:

Step 1: Click a row from the table, and it will display its information in the fields above

Step 2: Update the information

Step 3: Click the Update Transaction button

Step 4: Confirm the notification

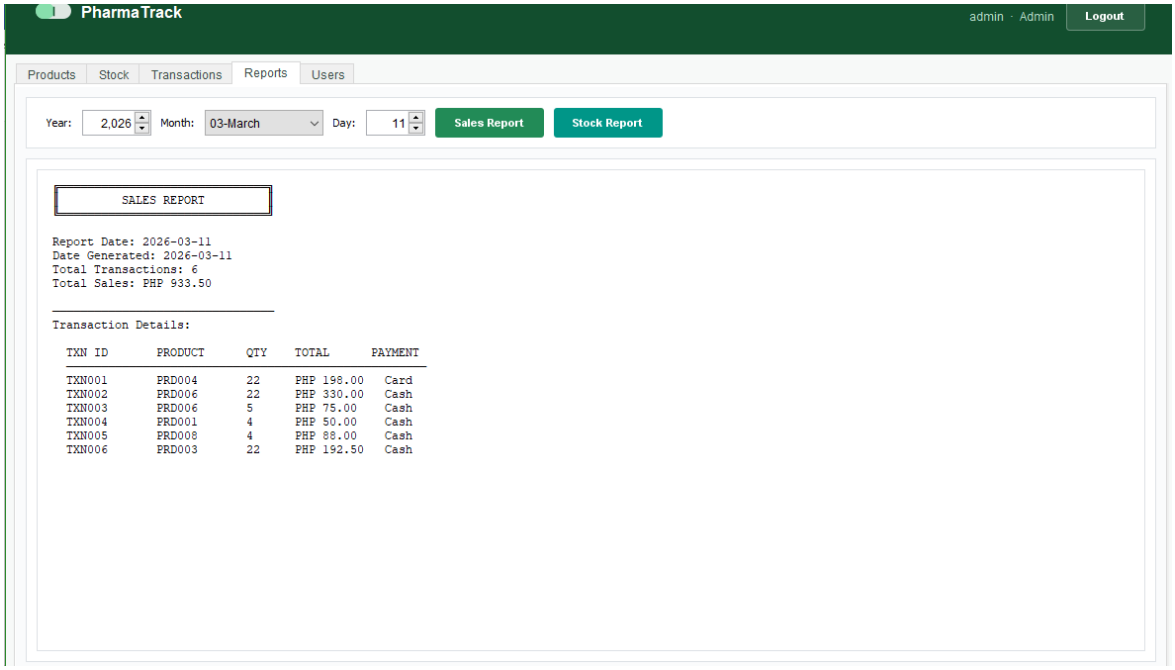
To exit the system, press X button or simply press Alt+f4.

To minimize the system, press the – button.

To return to Login Screen, click the Logout button.

<u>Features for Admins only:</u> **a.) Menu**

i.) Daily Report Management



To view the daily report:

Step 1: Choose the date that the admin will want to generate a report

Click the Sales Report button to generate only the sales

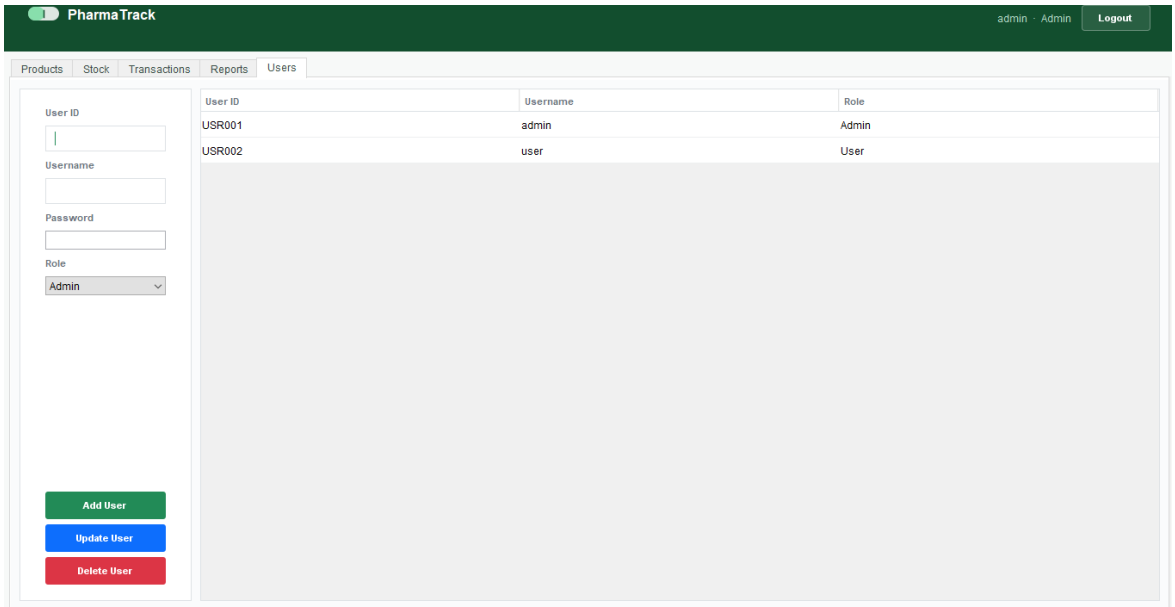
Click the Stock Report button to generate only the stock

To exit the system, press X button or simply press Alt+f4.

To minimize the system, press the – button.

To return to Login Screen, click the Logout button.

ii.) User Management



To add user/admin account:

Step 1: Fill in the User ID, Username, Password and Role

Step 2: Click the Add User button

Step 3: Confirm the notification

To update a user/admin information:

Step 1: Click a row from the table, and it will display its information in the fields above

Step 2: Update the information

Step 3: Click the Update User button

Step 4: Confirm the notification

To delete a user/admin account:

Step 1: Click a row from the table, and it will display its information in the fields above

Step 2: Click the Delete User button

Step 3: Confirm the notification

To exit the system, press X button or simply press Alt+f4.

To minimize the system, press the – button.

To return to Login Screen, click the Logout button.

SECTION 3 Contact Us

Call us at:

Cellphone Numbers:

- 09100558543 (TNT)

or send us an Email at:

- Shanallenb@gmail.com

Shan Allen Dayon

Purok10 Nacilla Village Bugac Maa DavaoCity
Contact no: 09100558543
Shanallenb@gmail.com

COMPUTER SKILLS

Computer Applications

- Microsoft Office (Word, Excel, and PowerPoint)

Programming Languages

- Java for Desktop Applications Development (NetBeans IDE)
- HTML5, JS, and CSS3

EDUCATIONAL BACKGROUND

Philippine Women's College of Davao
2024-Present
University Avenue, Juna Subdivision, Matina, Davao City
Senior High School

Maa national Highschool
2020-2024
Ma- a
Junior High School

Kapitan Tomas Monteverde sr central elementary school
Bangoy St, Poblacion District, Davao City,

SEMINARS/WORKSHOPS

ICT Workshop: Learning Android Studio Development using Java August 2025

ORGANIZATION/CLUB

Badminton Club

GLOSSARY

PharmaSync – A Pharmacy Inventory and Transaction Management System designed to improve pharmacy operations by automating tasks such as monitoring medicine stocks and handling sales transactions.

Inventory Management – The procedure of tracking, organizing, and maintaining the stock levels of medicines within the pharmacy.

Transaction – A documented record of a medicine sale that includes information such as the product name, quantity purchased, price, and transaction date.

Product (Medicine) – Any pharmaceutical item available for purchase in the pharmacy, including branded medicines, generic drugs, and over-the-counter products.

Stock – The available amount of a particular product currently stored in the pharmacy inventory.

Stock In – The process of increasing inventory by adding newly delivered medicines or additional quantities of existing products.

Stock Out – The decrease in the quantity of products in the inventory caused by sales or removal from stock.

Expiration Date – The specified date indicating when a medicine should no longer be used because it may not be safe or effective.

Sales Report – A system-generated document summarizing all completed sales transactions within a chosen time frame.

Stock Report – A report that displays the current condition of the inventory, including the available stock levels and products that are running low.

Daily Report – A compiled record that presents the pharmacy's daily sales activities, transactions, and inventory updates.

User – An individual who operates the system and has access to basic functions such as managing products, monitoring stock, and processing transactions.

Admin – A system administrator who has complete access to system functions, including user management, report generation, and overall control of system data.